

Assignment Case Study: Real-Time Emotion Detection Using Supervised Learning



Real-Time Emotion Detection Using Supervised Learning



Assignment Case Study

1. INTRODUCTION

This assignment aims to build a real-time emotion detection system using the concept of supervised machine learning. You will train a model to classify human emotions such as happy, sad, and neutral, then test the model in real time with your computer's webcam or your Raspberry Pi camera.



2. OBJECTIVE



To build a real-time emotion detection system using supervised machine learning techniques.



To install the required libraries:

```
pip install opencv-python scikit-learn numpy pandas matplotlib
```

3. TOOLS & LIBRARIES



OpenCV
For capturing real-time video and detecting faces.



NumPy & Pandas
For data handling and numerical operations.



Scikit-learn
For training and testing supervised learning models.



Matplotlib
For visualizing data and accuracy results.

4. WORKFLOW OVERVIEW



5. SUPERVISED LEARNING MODELS (Examples)



6. REAL-TIME EMOTION DETECTION

- 1 Capture video frames using OpenCV.
- 2 Detect faces using Haar Cascade Classifier.
- 3 Extract the face region.
- 4 Generate or extract feature values (e.g., eye openness, smile intensity).
- 5 Use the trained model to predict emotion.
- 6 Display the predicted emotion label on the video feed.
- 7 Press "Q" key to quit the program.



7. ASSIGNMENT STEPS

- Step 1 Import the Libraries**
Import all required Python libraries.
- Step 2 Prepare the Data**
Create or simulate a dataset with features (e.g., eye openness, smile intensity) and labels (happy, sad, neutral). Split into training and testing sets.
- Step 3 Train the Model**
Train a supervised learning model and evaluate its accuracy on the test data.
- Step 4 Real-Time Detection**
Connect the model to webcam, detect faces, predict emotions, and display results in real time.
- Step 5 Evaluate & Discuss**
Analyze results, discuss challenges, and suggest improvements.

8. DELIVERABLES



1. Python File
Submit your full code as 'emotion_detection.py'.



2. Screenshots / Video
Provide screenshots or a video showing your system in action.



3. Written Report
Include methodology, results, observations, challenges, and improvements.

9. EVALUATION CRITERIA

Criteria	Weight
Dataset Preparation	15%
Model Training	20%
Model Evaluation	15%
Real-Time Integration	25%
Report Quality	15%
Code Structure & Documentation	10%
Total	100%



10. EXPECTED LEARNING OUTCOMES

- ✓ Understand the concept of supervised learning.
- ✓ Train and evaluate classification models using Scikit-learn.
- ✓ Integrate a trained model into a real-time system using OpenCV.
- ✓ Analyze model performance and discuss ways to improve it.



11. OPTIONAL EXTENSIONS

- Use real dataset like FER2013.
- Use CNN for higher accuracy.
- Add more emotion categories.
- Use facial landmarks for real feature extraction.
- Deploy on Raspberry Pi.



Course Topic

Machine Learning – Supervised Learning

Assignment Title

Real-Time Emotion Detection Using Supervised Learning

1. Introduction

Artificial Intelligence (AI) systems are increasingly being used to understand and respond to human emotions. Emotion recognition has applications in healthcare, education, customer service, security systems, and human-computer interaction.

In this assignment, students will develop a real-time emotion detection system using supervised machine learning techniques. The system will classify human emotions such as **Happy**, **Sad**, and **Neutral** based on facial features extracted from webcam images.

The project combines machine learning model development with **real-time computer vision using OpenCV**.

2. Learning Objectives

Upon successful completion of this assignment, students will be able to:

- Understand the principles of supervised learning.
- Create and prepare datasets for classification tasks.
- Train and evaluate machine learning classification models.
- Apply machine learning models to real-time applications.
- Use OpenCV for video capture and face detection.
- Analyze model performance and identify limitations.
- Propose improvements for future AI systems.

3. Case Scenario

A technology company wants to build an intelligent system capable of identifying a user's emotional state through a **Raspberry Pi camera feed**.

The system should:

1. Detect a person's face in real time.
2. Extract facial features relevant to emotion recognition.
3. Classify the detected emotion.

4. Display the predicted emotion on the screen.

As a **machine learning engineer**, your task is to design, implement, and evaluate a prototype of this emotion detection system.

4. Background Theory

Supervised Learning: Supervised learning is a machine learning approach where **models learn from labeled data**.

Each training example contains:

- Input features
- Correct output label

The model learns the relationship between inputs and outputs and then predicts labels for unseen data.

Example

Eye Openness	Smile Intensity	Emotion
0.8	0.9	Happy
0.3	0.2	Sad
0.5	0.4	Neutral

The goal is for the model to learn patterns that distinguish these emotional states.

5. Required Tools and Libraries

Students must use Python and the following libraries:

Library	Purpose
OpenCV	Webcam access and face detection
NumPy	Numerical operations
Pandas	Data handling
Scikit-learn	Machine learning models
Matplotlib	Data visualization

Installation

```
pip install opencv-python scikit-learn numpy pandas matplotlib
```

6. Assignment Tasks

Task 1: Import Required Libraries

Import all necessary Python libraries required for:

- Data processing
 - Model training
 - Model evaluation
 - Real-time Raspberry Pi access
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Task 2: Dataset Preparation

Create a simulated dataset containing facial-expression-related features.

Example features:

- Eye openness
- Smile intensity

Each sample must belong to one of the following classes:

- Happy
- Sad
- Neutral

Requirements

- Generate sufficient training samples.
- Store the data in a Pandas DataFrame.
- Split the dataset into:
 - Training set
 - Testing set

Suggested split:

- 80% Training
 - 20% Testing
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Task 3: Train a Supervised Learning Model

Select one supervised learning algorithm:

- Option A: Logistic Regression
- Option B: Support Vector Machine (SVM)
- Option C: Random Forest Classifier

Requirements

- Train the model using the training dataset.
- Evaluate performance using the testing dataset.
- Calculate classification accuracy.

Suggested evaluation metrics:

- Accuracy
 - Precision
 - Recall
 - F1-score
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Task 4: Real-Time Emotion Detection

Connect the trained model to a webcam.

Real-Time Workflow

1. Capture video frames using OpenCV.
2. Detect faces using Haar Cascade Classifier.
3. Extract the face region.
4. Generate or estimate feature values.
5. Feed features into the trained model.
6. Predict emotion.
7. Display the predicted emotion label on the video feed.

User Interaction

The application should continue running until the user presses **Q** to quit the program. /

Task 5: Evaluation and Discussion

After completing the implementation, evaluate the system.

Discuss:

Model Performance

- What accuracy did the model achieve?
- Were predictions consistent?

Real-Time Performance

- Did the system respond quickly?
- Were emotions correctly identified?

Challenges Encountered

Possible issues:

- Poor lighting
- Camera quality
- Incorrect face detection
- Limited training data
- False predictions

Future Improvements

Examples:

- Larger datasets
- Deep learning models
- More advanced facial features
- Real facial landmark extraction
- Better preprocessing techniques

7. Optional Advanced Extension

Students seeking additional challenges may implement one or more of the following enhancements:

Real Dataset

Replace the simulated dataset with:

- FER2013 (Facial Expression Recognition Dataset)

Deep Learning

Use a Convolutional Neural Network (CNN) instead of traditional machine learning models.

Additional Emotions

Add:

- Angry

- Fearful
- Surprised
- Disgusted